

Please solve the following exercises at home prior to the tutorial:

Exercise 1

Solve the following questions by first converting the decimal numbers to binary and then performing the addition in binary.

- a) $15_{10} + 31_{10} = ?_2$
- b) $105_{10} + 21_{10} = ?_2$

The following exercises will be done during the tutorial:

Exercise 2

Subtract the following numbers in two's complement, base 2, using 8 digits.

- a) $43_{10} - 11_{10}$
- b) $17_{10} - 109_{10}$

Exercise 3

- a) What is the largest unsigned 5-bit number?
- b) What is the largest 5-bit number in two's complement?
- c) What is the smallest 5-bit number in two's complement?

Exercise 4

Unix operating systems count the time since 1.1.1970 by incrementing a number every second. What is the resulting problem when using a 32-bit signed integer? When will this problem manifest?

Exercise 5

Subtract the following numbers using 10-Complement, base 10, with 5 digits:

- a) $25737_{10} - 18547_{10}$
- b) $2737_{10} - 4578_{10}$



Exercise 6

Subtract the following numbers using Ones'-Complement in base 2 with 8 digits:

$$57_{10} - 122_{10}$$

Exercise 7

Give the B-Complement (B=16 or B=5) for the following numbers:

- a) $AFFE_{16}$
- b) 124_5

Exercise 8

Subtract the following numbers using 9-Complement, base 10, with 5 digits:

- a) $25737_{10} - 18547_{10}$
- b) $2737_{10} - 4578_{10}$